Seventh Grade Tile Task

Standard 7.RP.1

7

1. a) 24 miles per 8 hours: unit rate? 3

b) 12 miles per 2 hours: unit rate? 6

c) 14 miles per 2 hours: unit rate? 7

2. a) 12 miles per 6 hours: unit rate? 2

b) 72 miles per 9 hours: unit rate? 8

c) 45 miles per 5 hours: unit rate? 9

3. a) 20 miles per 4 hours: unit rate? 5

b) 16 miles per 4 hours: unit rate? 4

c) 4 miles per 4 hours: unit rate? 1

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.RP.2

7

1. a) y = 9x. If x = 1, what is y? 9

b) y = 3x. If x = 1, what is y? 3

c) y = 1x. If x = 1, what is y? 1

2. a) y = 1x. If x = 2, what is y? 2

b) y = 1x. If x = 4, what is y? 4

c) y = 1x. If x = 5, what is y? 5

3. a) y = 1x. If x = 7, what is y? 7

b) y = 1x. If x = 8, what is y? 8

c) y = 1x. If x = 6, what is y? 6

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.RP.3

7

1. a) 10% of 20 — 2

b) 10% of 40 — 4

c) 10% of 80 — 8

2. a) 50% of 6 — 3

b) 50% of 10 — 5

c) 25% of 24 — 6

3. a) 10% of 70 — 7

b) 10% of 10 — 1

c) 25% of 36 — 9

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: SEVENTH GRADE

Seventh Grade Tile Task

Standard 7.NS.1

7

1. **a)** -9 + 2 = ? (distance from 0) **7**

   b) -2 + 2 = ? (distance from 0) **0**

   c) -7 + 4 = ? (distance from 0) **3**

2. **a)** -0 + 8 = ? (distance from 0) **8**

   b) -0 + 2 = ? (distance from 0) **2**

   c) -0 + 6 = ? (distance from 0) **6**

3. **a)** -3 + 8 = ? (distance from 0) **5**

   b) -8 + 7 = ? (distance from 0) **1**

   c) -1 + 5 = ? (distance from 0) **4**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.NS.2

7

1. **a)** Absolute value of (-4) × 1 — 4

   b) Absolute value of (-8) × 1 — 8

   c) Absolute value of (-1) × 5 — 5

2. **a)** Absolute value of (-1) × 1 — 1

   b) Absolute value of (-9) × 1 — 9

   c) Absolute value of (-7) × 1 — 7

3. **a)** Absolute value of (-1) × 3 — 3

   b) Absolute value of (-2) × 1 — 2

   c) Absolute value of (-3) × 2 — 6

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.NS.3

7

1. **a)** Distance between -6 and 3 on a number line — 9
   b) Distance between -1 and 7 on a number line — 8
   c) Distance between -1 and 1 on a number line — 2

2. **a)** Distance between -0 and 1 on a number line — 1
   b) Distance between -3 and 0 on a number line — 3
   c) Distance between -2 and 4 on a number line — 6

3. **a)** Distance between -4 and 1 on a number line — 5
   b) Distance between -1 and 6 on a number line — 7
   c) Distance between -3 and 1 on a number line — 4

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.EE.1

7

1. a) Coefficient of x in 2(x + 1) — 2

b) Coefficient of x in 7(x + 1) — 7

c) Coefficient of x in 5(x + 1) — 5

2. a) Constant term of 4(x + 1) — 4

b) Coefficient of x in 9(x + 1) — 9

c) Constant term of 1(x + 8) — 8

3. a) Coefficient of x in 1(x + 1) — 1

b) Constant term of 3(x + 1) — 3

c) Constant term of 3(x + 2) — 6

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.EE.2

7

1. a) Coefficient of x in 7x + 0 — 7

b) Coefficient of x in 2x + 4 — 2

c) Coefficient of x in 3x + 6 — 3

2. a) Coefficient of x in 9x + 2 — 9

b) Coefficient of x in 5x + 9 — 5

c) Coefficient of x in 4x + 1 — 4

3. a) Coefficient of x in 6x + 8 — 6

b) Coefficient of x in 8x + 9 — 8

c) Coefficient of x in 1x + 5 — 1

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.EE.3

7

1. a) Value of 3x + 2 when x = 1 — **5**

b) Value of 3x + 0 when x = 0 — **0**

c) Value of 1x + 8 when x = 1 — **9**

2. a) Value of 1x + 4 when x = 4 — **8**

b) Value of 1x + 0 when x = 4 — **4**

c) Value of 2x + 1 when x = 1 — **3**

3. a) Value of 1x + 0 when x = 2 — **2**

b) Value of 1x + 6 when x = 1 — **7**

c) Value of 3x + 3 when x = 1 — **6**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.EE.4

7

1. a) Solve 3x + 4 = 10: x = ? **2**

b) Solve 6x + 5 = 29: x = ? **4**

c) Solve 3x + 5 = 5: x = ? **0**

2. a) Solve 5x + 9 = 54: x = ? **9**

b) Solve 8x + 3 = 27: x = ? **3**

c) Solve 8x + 2 = 50: x = ? **6**

3. a) Solve 8x + 1 = 9: x = ? **1**

b) Solve 7x + 2 = 51: x = ? **7**

c) Solve 9x + 4 = 49: x = ? **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.G.1

7

1. a) Scale 1 cm : 8 cm, a 1 cm length is how many cm real? **8**

b) Scale 1 cm : 5 cm, a 1 cm length is how many cm real? **5**

c) Scale 1 cm : 9 cm, a 1 cm length is how many cm real? **9**

2. a) Scale 1 cm : 7 cm, a 1 cm length is how many cm real? **7**

b) Scale 1 cm : 2 cm, a 2 cm length is how many cm real? **4**

c) Scale 1 cm : 3 cm, a 1 cm length is how many cm real? **3**

3. a) Scale 1 cm : 2 cm, a 1 cm length is how many cm real? **2**

b) Scale 1 cm : 1 cm, a 1 cm length is how many cm real? **1**

c) Scale 1 cm : 2 cm, a 3 cm length is how many cm real? **6**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.G.2

7

1. a) Missing angle of a triangle with angles 82° and 91° — 7

b) Missing angle of a triangle with angles 85° and 89° — 6

c) Missing angle of a triangle with angles 94° and 81° — 5

2. a) Missing angle of a triangle with angles 95° and 81° — 4

b) Missing angle of a triangle with angles 81° and 90° — 9

c) Missing angle of a triangle with angles 97° and 80° — 3

3. a) Missing angle of a triangle with angles 95° and 84° — 1

b) Missing angle of a triangle with angles 86° and 86° — 8

c) Missing angle of a triangle with angles 92° and 86° — 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.G.3

7

1. **a)** Sides of an octagon cross section of a cube? **8**

   b) Sides of a hexagon cross section of a cube? **6**

   c) Sides of a heptagon cross section of a prism? **7**

2. **a)** Sides of a triangle cross section of a prism? **3**

   b) Sides of a nonagon cross section of a prism? **9**

   c) Sides of a square cross section of a cube? **4**

3. **a)** Bases of a cylinder? **2**

   b) Sides of a triangle cross section of a cone? **0**

   c) Bases of a cone? **1**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.G.4

7

1. a) Radius of a circle with diameter 4 — 2

b) Radius of a circle with diameter 10 — 5

c) Radius of a circle with diameter 2 — 1

2. a) Radius of a circle with diameter 8 — 4

b) Using π ≈ 3, circumference of diameter 3 — 9

c) Radius of a circle with diameter 16 — 8

3. a) Using π ≈ 3, circumference of diameter 1 — 3

b) Using π ≈ 3, circumference of diameter 2 — 6

c) Radius of a circle with diameter 14 — 7

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.G.5

7

1. **a)** Complementary angle of 89° — 1
   b) Complementary angle of 88° — 2
   c) Complementary angle of 81° — 9
2. **a)** Complementary angle of 82° — 8
   b) Supplementary angle of 175° — 5
   c) Complementary angle of 83° — 7
3. **a)** Supplementary angle of 177° — 3
   b) Complementary angle of 84° — 6
   c) Supplementary angle of 176° — 4

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.G.6

7

1. **a)** Area of a 2 by 1 rectangle — 2

   b) Area of a 3 by 1 rectangle — 3

   c) Area of a 1 by 4 rectangle — 4

2. **a)** Area of a 1 by 6 rectangle — 6

   b) Area of a 4 by 2 rectangle — 8

   c) Area of a 1 by 1 rectangle — 1

3. **a)** Area of a 1 by 5 rectangle — 5

   b) Area of a 3 by 3 rectangle — 9

   c) Area of a 1 by 7 rectangle — 7

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.SP.1

7

1. **a)** How many items in a sample of 2? 2

   b) How many items in a sample of 8? 8

   c) How many items in a sample of 3? 3

2. **a)** How many items in a sample of 5? 5

   b) How many items in a sample of 9? 9

   c) How many items in a sample of 4? 4

3. **a)** How many items in a sample of 1? 1

   b) How many items in a sample of 6? 6

   c) How many items in a sample of 7? 7

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.SP.2

7

1. a) If 5 out of 10 are red, how many of 10 are red? **5**

b) If 9 out of 10 are red, how many of 10 are red? **9**

c) If 6 out of 10 are red, how many of 10 are red? **6**

2. a) If 8 out of 10 are red, how many of 10 are red? **8**

b) If 1 out of 10 are red, how many of 10 are red? **1**

c) If 2 out of 10 are red, how many of 10 are red? **2**

3. a) If 0 out of 10 are red, how many of 10 are red? **0**

b) If 3 out of 10 are red, how many of 10 are red? **3**

c) If 7 out of 10 are red, how many of 10 are red? **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.SP.3

7

1. a) Group A mean 9, Group B mean 5: how much more? — **4**

b) Group A mean 7, Group B mean 1: how much more? — **6**

c) Group A mean 9, Group B mean 0: how much more? — **9**

2. a) Group A mean 8, Group B mean 5: how much more? — **3**

b) Group A mean 7, Group B mean 6: how much more? — **1**

c) Group A mean 6, Group B mean 1: how much more? — **5**

3. a) Group A mean 8, Group B mean 1: how much more? — **7**

b) Group A mean 8, Group B mean 0: how much more? — **8**

c) Group A mean 2, Group B mean 0: how much more? — **2**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.SP.4

7

1. a) Mean of 3 and 13 — 8

b) Mean of 5 and 13 — 9

c) Mean of 1 and 1 — 1

2. a) Mean of 3 and 1 — 2

b) Mean of 3 and 3 — 3

c) Mean of 3 and 11 — 7

3. a) Mean of 0 and 0 — 0

b) Mean of 3 and 5 — 4

c) Mean of 5 and 5 — 5

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.SP.5

7

1. **a)** Chance of 1 out of 10: build the numerator — 1
   b) Chance of 8 out of 10: build the numerator — 8
   c) Chance of 5 out of 10: build the numerator — 5

2. **a)** Chance of 2 out of 10: build the numerator — 2
   b) Chance of 6 out of 10: build the numerator — 6
   c) Chance of 9 out of 10: build the numerator — 9

3. **a)** Chance of 0 out of 10: build the numerator — 0
   b) Chance of 4 out of 10: build the numerator — 4
   c) Chance of 3 out of 10: build the numerator — 3

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.SP.6

7

1. **a)** Heads in 3 of 10 flips: build the number 3

   b) Heads in 9 of 10 flips: build the number 9

   c) Heads in 4 of 10 flips: build the number 4

2. **a)** Heads in 5 of 10 flips: build the number 5

   b) Heads in 8 of 10 flips: build the number 8

   c) Heads in 7 of 10 flips: build the number 7

3. **a)** Heads in 6 of 10 flips: build the number 6

   b) Heads in 1 of 10 flips: build the number 1

   c) Heads in 2 of 10 flips: build the number 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.SP.7

7

1. **a)** Spinner with 7 equal parts: how many outcomes? **7**
   b) Spinner with 3 equal parts: how many outcomes? **3**
   c) Spinner with 5 equal parts: how many outcomes? **5**

2. **a)** Spinner with 9 equal parts: how many outcomes? **9**
   b) Spinner with 1 equal parts: how many outcomes? **1**
   c) Spinner with 2 equal parts: how many outcomes? **2**

3. **a)** Spinner with 6 equal parts: how many outcomes? **6**
   b) Spinner with 4 equal parts: how many outcomes? **4**
   c) Spinner with 8 equal parts: how many outcomes? **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Seventh Grade Tile Task

Standard 7.SP.8

7

1. **a)** Outcomes of a spinner with 4 parts and one with 2 parts — 8
   b) Outcomes of a spinner with 3 parts and one with 1 parts — 3
   c) Outcomes of a spinner with 6 parts and one with 1 parts — 6

2. **a)** Outcomes of a spinner with 1 parts and one with 5 parts — 5
   b) Outcomes of a spinner with 2 parts and one with 1 parts — 2
   c) Outcomes of a spinner with 2 parts and one with 2 parts — 4

3. **a)** Outcomes of a spinner with 9 parts and one with 1 parts — 9
   b) Outcomes of a spinner with 1 parts and one with 7 parts — 7
   c) Outcomes of a spinner with 1 parts and one with 1 parts — 1

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9